

Deliverable 2 Report

Deliverable 2 was around writing Python code for data organization, creating chips for the model to make inferences on, and being able to segment changes by change type, location, and time period. For this deliverable, a more efficient setup of the HDF5 tile files was created, which changed the data chunks to be for 256x256 chips with multiple timesteps instead of row ordered pixels. From this setup, I created blocks of 4x4 chips (~1km x ~1km) with a ring of extra data for buffer. These blocks can be easily distributed to different CPUs, to parallelize the processing of an entire tile.

Each CPU first creates a before and after pair for the 16 chips for each change date specified by the user. It then handles making predictions on all the chip pairs for all change dates. In order to make more reliable inferences, the chip pairs are ran multiple times through the model, with vertical, horizontal, or diagonal shifts added each time. In total, for each change date there are 81 chip pair inferences yielded (16 original, 20 horizontally shifted, 20 vertically shifted, 25 diagonally shifted). With these results, a filter is applied to only keep pixels where change was detected multiple times.

The pixels that pass the filter are used at the block level to create aggregated patches of changes using morphological closing. These patches are then converted into polygons and then passed to a master CPU. The master CPU does an additional process of aggregation between all the block patches for the entire tile before creating a final full tile file with the polygons where changes were inferred to have happened.

The code for this deliverable can be found on the GitHub project at the link below.

Project GitHub link - https://github.com/S2change/vegetation_loss